

The Unified Pion Field Architecture: Simulating Hadronic Benevolence via Yukawa Binding

Version 2.0 — Revised & Scientifically Deepened | June 2026

Classification: Tardigradia Game Engine | Physics Deep Dive Series | TGPU v2.0 **Particle Class:** Pion π | Masses: $\pi^\pm=139.6$ MeV, $\pi^0=135.0$ MeV | Spin: 0 | Pseudo-Goldstone boson (chiral symmetry) | Yukawa carrier **Source:** Extends v1.0Pion Deep Dive_ One-Pion Universe.pdf (V1.0)

Δ Change Log V1.0 $\square$ V2.0

Parameter	V1.0	V2.0	Source
Pion form factor (CLAS12)	General pion form factor	V2.0: CLAS12@JLab 2024: pion electromagnetic form factor F_π measured at $Q^2=0.6\text{-}2.5$ GeV^2 . $F_\pi(Q^2)$ crossover from soft to hard scattering regime confirmed. V2.0: implements both soft (VMD) and hard (pQCD) form factor regimes with smooth interpolation at $Q^2\sim 1.5$ GeV^2	JLab CLAS12 2024
Chiral perturbation theory update	Chiral symmetry breaking mentioned	V2.0: FLAG 2024 lattice average: $m_u=2.16$ MeV, $m_d=4.67$ MeV, $m_\pi = \sqrt{B(m_u+m_d)}$ where B is chiral condensate. V2.0: pion mass dynamically computed from quark masses — changes when quark mass parameters vary in simulation	FLAG 2024; ChPT NLO
Pion decay constant F_π	Chiral symmetry	V2.0: $F_\pi = 92.07\pm0.46$ MeV (FLAG 2024 lattice). Critical for chiral effective theory. In simulation: pion annihilation rate $\text{Gamma}(\pi^0\rightarrow\gamma\gamma) = \alpha^2 m_\pi^3 / (64\pi^3 F_\pi^2)$. Updated from V1.0	FLAG 2024 F_π

Parameter	V1.0	V2.0	Source
Pionium bound state	Not covered	V2.0: DIRAC experiment (CERN): pionium ($\pi^+\pi^-$ bound state) lifetime measured $\tau = (2.91 \pm 0.17) \times 10^{-15}$ s. V2.0 adds PIONIUM = bound state of two pion worldline intersections. Lifetime encodes pi-pi scattering lengths a_0^0 , a_0^2	DIRAC 2022 updated PDG 2024

1. The One-Particle Universe Framework — V2.0 Update

The foundational architecture established in V1.0 — Worldline Monism, the “Origami Universe” metaphor, and the Object-Oriented ontology distinguishing intrinsic from extrinsic properties — is **fully preserved** and remains physically rigorous. V2.0 layers NEW discoveries from 2022-2026 atop this framework.

1.1 WebGPU Compute Architecture (V2.0 Core Infrastructure)

V1.0 designed for WebGL/Three.js on the CPU-side particle update loop. V2.0 upgrades to WebGPU (available in Chrome 113+ May 2023, Firefox 2024):

```
// WGSL Compute Shader - V2.0 Particle Integration
@binding(0) @group(0) var<storage, read_write> particles: array<ParticleState>;

struct ParticleState {
    position: vec4f,
    momentum: vec4f,
    spin: vec2f,           // Spinor components
    berry_phase: f32,      // Geometric phase accumulated
    proper_time: f32,      // tau - worldline parameter
    color_charge: u32,      // Packed color in 2 bits (R=0,G=1,B=2)
    quantum_numbers: u32,  // Packed: isospin, strangeness, charm, beauty
    decay_lifetime: f32,
    entanglement_id: i32,  // -1 = unentangled; >=0 = entanglement group
}

@compute @workgroup_size(64)
fn updateParticles(@builtin(global_invocation_id) id: vec3u) {
    let i = id.x;
    if (i >= arrayLength(&particles)) { return; }

    var p = particles[i];

    // Feynman-Schwinger proper time evolution
    let dtau = uniforms.dt / p.mass; // proper time step
```

```

// Update worldline parameter
p.proper_time += dtau;

// Berry phase accumulation (topological phase)
p.berry_phase += dot(uniforms.gauge_field, p.momentum.xyz) * dtau;

// Lorentz force (for charged particles)
let F = cross(uniforms.B_field, p.momentum.xyz) * p.charge_sign;
p.momentum.xyz += F * uniforms.dt;
p.position.xyz += p.momentum.xyz * uniforms.dt / p.energy;

particles[i] = p;
}

```

1.2 Updated Physical Constants (CODATA 2022)

Constant	V1.0 Value	V2.0 Value (CODATA 2022)
m _e	9.10938356×10 ⁻³¹ kg	9.1093837139×10 ⁻³¹ kg
e	1.60217663×10 ⁻¹⁹ C	1.602176634×10 ⁻¹⁹ C (exact)
ħ	1.054571817×10 ⁻³⁴ J·s	1.054571817×10 ⁻³⁴ J·s (exact)
c	2.99792458×10 ⁸ m/s	2.99792458×10 ⁸ m/s (exact)
α	1/137.036	1/137.035999084 ±0.000000021

2. V2.0 Enhanced Data Structures

Extending V1.0 struct definitions with 2022-2026 discoveries:

```

// V2.0 Universal Particle State – extends V1.0
struct ParticleInstanceV2 {
    // Inherited from V1.0
    double proper_time;           // tau worldline parameter (meter)
    glm::dvec3 position;          // 3-space vector
    glm::dvec3 momentum;          // 3-momentum (relativistic)
    float spin_z;                 // +0.5 or -0.5 (or eigenvalue)
    float berry_phase;            // Accumulated Berry/Aharonov-Bohm phase

    // V2.0 ADDITIONS
    float entanglement_r;         // Reduced density matrix purity 0-1
    int entanglement_partner_id;  // -1 = none; global particle ID
    float flavor_oscillation_phase; // For neutrinos/kaons
    uint32_t color_bits;          // 3-bit: R(b0),G(b1),B(b2)
    float decay_probability_dt;    // P(decay in dt) for unstable particles
    float vacuum_alignment;       // Order parameter for condensate mode

    // WebGPU Buffer packing
    // Total: ~100 bytes/particle * 10^6 particles = 100 MB GPU memory
    float _pad[2];                // Align to 16 bytes for WGSL
};

```

8. V2.0 Benevolence Metric Update

The “Benevolence” concept central to all OEU documents is given a precise V2.0 mathematical definition grounded in 2022-2026 quantum information theory:

Benevolence = Quantum Coherence + Information Preservation + Symmetry Maintenance

$$B(t) = C(t) \cdot F(t) \cdot S(t)$$

Where: - $C(t)$ = l_1 -norm coherence of particle density matrix (Baumgratz et al. 2014) - $F(t)$ = quantum Fisher information / classical Fisher information (ratio ≥ 1) - $S(t)$ = fraction of symmetries unbroken at time t

V2.0 game mechanic: Benevolence score increases when: - Particles maintain long entanglement coherence (high C) - Topological protections are active (Berry phase winding conserved) - Symmetry groups are preserved in interactions (no spurious symmetry breaking)

*End of v1.0Pion Deep Dive_ One-Pion Universe V2.0 | Tardigradia Game Engine SubParticles Series
This document supersedes V1.0 in all physics specifications.*